DECISION TREE CART ALGORITHM

Using Gini Index

Example 1:

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	−
4	F	F	4.0	+
5	F	T	7.0	−
6	F	T	3.0	−
7	F	F	8.0	−
8	T	F	7.0	+
9	F	T	5.0	−

Compute the Gini Index of the attributes a1.

$$Gini = 1 - \sum_{i=1}^{n} (p_i)^2$$

$$Gini(T) = 1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2 = 0.375$$

$$Gini(F) = 1 - \left(\frac{1}{5}\right)^2 - \left(\frac{4}{5}\right)^2 = 0.32$$

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	−
4	F	F	4.0	+
5	F	T	7.0	−
6	F	T	3.0	−
7	F	F	8.0	−
8	T	F	7.0	+
9	F	T	5.0	−

Compute the Gini Index of the attributes a1.

$$GiniIndex(a_1) = \sum_{v \in \{T,F\}} \frac{|S_v|}{|S|} Gini(S_v)$$

$$GiniIndex(a_1) = \left(\frac{4}{9}\right) * Gini(T) + \left(\frac{5}{9}\right) * Gini(F)$$

$$GiniIndex(a_1) = \left(\frac{4}{9}\right) * 0.375 + \left(\frac{5}{9}\right) * 0.32$$

$$GiniIndex(a_1) = 0.3444$$

Compute the Gini Index of the attributes a2.

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	−
4	F	F	4.0	+
5	F	T	7.0	−
6	F	T	3.0	−
7	F	F	8.0	−
8	T	F	7.0	+
9	F	T	5.0	−

$$Gini = 1 - \sum_{i=1}^{n} (p_i)^2$$

$$Gini(T) = 1 - \left(\frac{2}{5}\right)^2 - \left(\frac{3}{5}\right)^2 = 0.48$$

$$Gini(F) = 1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2 = 0.5$$

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	−
4	F	F	4.0	+
5	F	T	7.0	−
6	F	T	3.0	−
7	F	F	8.0	−
8	T	F	7.0	+
9	F	T	5.0	−

Compute the Gini Index of the attributes a2.

$$GiniIndex(a_2) = \sum_{v \in \{T,F\}} \frac{|S_v|}{|S|} Gini(S_v)$$

$$GiniIndex(a_2) = \left(\frac{5}{9}\right) * Gini(T) + \left(\frac{4}{9}\right) * Gini(F)$$

$$GiniIndex(a_2) = \left(\frac{5}{9}\right) * 0.48 + \left(\frac{4}{9}\right) * 0.5$$

$$GiniIndex(a_2) = 0.4889$$

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	−
4	F	F	4.0	+
5	F	T	7.0	−
6	F	T	3.0	−
7	F	F	8.0	−
8	T	F	7.0	+
9	F	T	5.0	−

Which is the best splitting attribute between a1 and a2.

$$\boldsymbol{GiniIndex(a_1) = 0.3444}$$

$$\boldsymbol{GiniIndex(a_2) = 0.4889}$$

Smaller GiniIndex Produces Better Split

Hence, attribute a1 is the best split attribute

Example 2:

Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Gini index

Gini index is a metric for classification tasks in CART. It stores sum of squared probabilities of each class. We can formulate it as illustrated below.

$$\text{Gini} = 1 - \Sigma\,(Pi)^2 \text{ for } i=1 \text{ to number of classes}$$

Outlook

sunny, overcast or rain.

Outlook	Yes	No	Number of instances
Sunny	2	3	5
Overcast	4	0	4
Rain	3	2	5

$$\text{Gini(Outlook=Sunny)} = 1 - (2/5)^2 - (3/5)^2 = 1 - 0.16 - 0.36 = 0.48$$

$$\text{Gini(Outlook=Overcast)} = 1 - (4/4)^2 - (0/4)^2 = 0$$

$$\text{Gini(Outlook=Rain)} = 1 - (3/5)^2 - (2/5)^2 = 1 - 0.36 - 0.16 = 0.48$$

calculate weighted sum of gini indexes for outlook feature.

$$\text{Gini(Outlook)} = (5/14) \times 0.48 + (4/14) \times 0 + (5/14) \times 0.48 = 0.171 + 0 + 0.171 = 0.342$$

Temperature

values: Cool, Hot Mild.

Temperature	Yes	No	Number of instances
Hot	2	2	4
Cool	3	1	4
Mild	4	2	6

$$Gini(Temp=Hot) = 1 - (2/4)^2 - (2/4)^2 = 0.5$$

$$Gini(Temp=Cool) = 1 - (3/4)^2 - (1/4)^2 = 1 - 0.5625 - 0.0625 = 0.375$$

$$Gini(Temp=Mild) = 1 - (4/6)^2 - (2/6)^2 = 1 - 0.444 - 0.111 = 0.445$$

calculate weighted sum of gini index for temperature feature

$$Gini(Temp) = (4/14) \times 0.5 + (4/14) \times 0.375 + (6/14) \times 0.445 = 0.142 + 0.107 + 0.190 = 0.439$$

Humidity

It can be high or normal.

Humidity	Yes	No	Number of instances
High	3	4	7
Normal	6	1	7

Gini(Humidity=High) = $1 - (3/7)^2 - (4/7)^2 = 1 - 0.183 - 0.326 = 0.489$

Gini(Humidity=Normal) = $1 - (6/7)^2 - (1/7)^2 = 1 - 0.734 - 0.02 = 0.244$

Weighted sum for humidity feature will be calculated next

Gini(Humidity) = $(7/14) \times 0.489 + (7/14) \times 0.244 = 0.367$

Wind

weak and strong.

Wind	Yes	No	Number of instances
Weak	6	2	8
Strong	3	3	6

$$Gini(Wind=Weak) = 1 - (6/8)^2 - (2/8)^2 = 1 - 0.5625 - 0.062 = 0.375$$

$$Gini(Wind=Strong) = 1 - (3/6)^2 - (3/6)^2 = 1 - 0.25 - 0.25 = 0.5$$

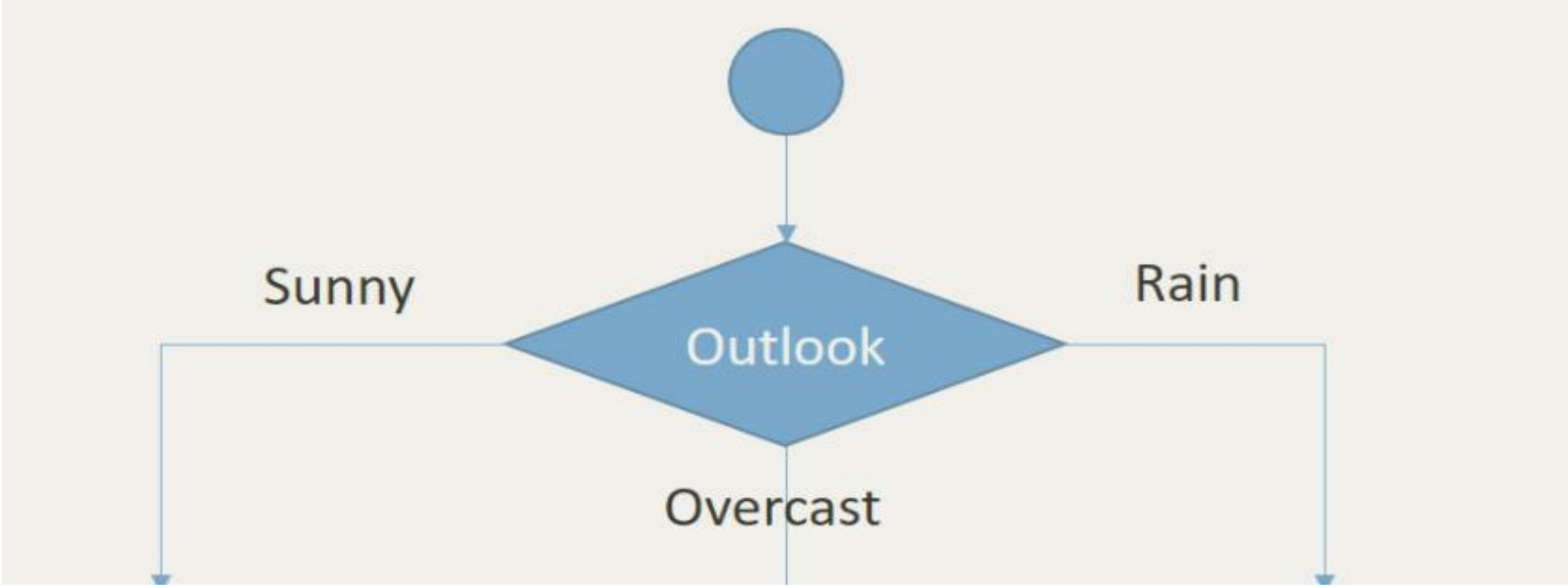
$$Gini(Wind) = (8/14) \times 0.375 + (6/14) \times 0.5 = 0.428$$

So which should be the root node?

The attribute having the smallest Gini Index value will be selected.

Feature	Gini index
Outlook	0.342
Temperature	0.439
Humidity	0.367
Wind	0.428

So it will be Outlook



sub dataset in the overcast leaf has only yes

This means that overcast leaf is over.

Now For Sunny: (Deciding Left branch node)

Day	Outlook	Temp.	Humidity	Wind	Decision
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes

Gini of temperature for sunny outlook

Temperature	Yes	No	Number of instances
Hot	0	2	2
Cool	1	0	1
Mild	1	1	2

Gini(Outlook=Sunny and Temp.=Hot) = $1 - (0/2)^2 - (2/2)^2 = 0$

Gini(Outlook=Sunny and Temp.=Cool) = $1 - (1/1)^2 - (0/1)^2 = 0$

Gini(Outlook=Sunny and Temp.=Mild) = $1 - (1/2)^2 - (1/2)^2 = 1 - 0.25 - 0.25 = 0.5$

Gini(Outlook=Sunny and Temp.) = $(2/5)x0 + (1/5)x0 + (2/5)x0.5 = 0.2$

Gini of humidity for sunny outlook

Humidity	Yes	No	Number of instances
High	0	3	3
Normal	2	0	2

Gini(Outlook=Sunny and Humidity=High) = $1 - (0/3)^2 - (3/3)^2 = 0$

Gini(Outlook=Sunny and Humidity=Normal) = $1 - (2/2)^2 - (0/2)^2 = 0$

Gini(Outlook=Sunny and Humidity) = $(3/5)x0 + (2/5)x0 = 0$

Gini of wind for sunny outlook

Wind	Yes	No	Number of instances
Weak	1	2	3
Strong	1	1	2

Gini(Outlook=Sunny and Wind=Weak) = $1 - (1/3)^2 - (2/3)^2 = 0.266$

Gini(Outlook=Sunny and Wind=Strong) = $1- (1/2)^2 - (1/2)^2 = 0.2$

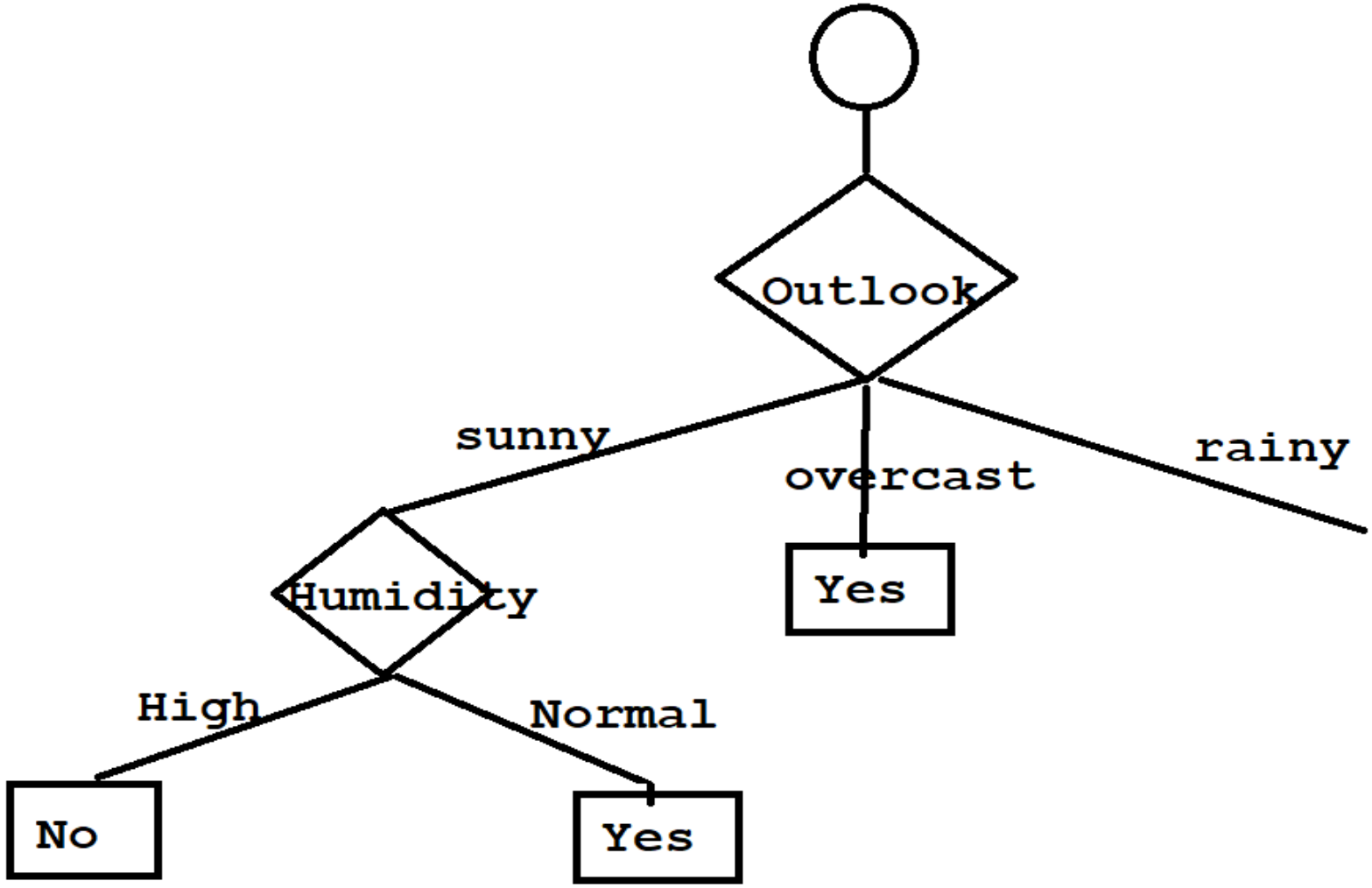
Gini(Outlook=Sunny and Wind) = $(3/5)\text{x}0.266 + (2/5)\text{x}0.2 = 0.466$

So which should be the node if Outlook is Sunny?

The attribute having the smallest Gini Index value will be selected.

Feature	Gini index
Temperature	0.2
Humidity	0
Wind	0.466

So Humidity will be selected as the node if outlook is sunny.



For Humidity there is no need to calculate again.
Since all humidity records with High as value, the decision is “No”
And all humidity records with Normal as value, the decision is “Yes”

Now For Rainy: (Deciding Right branch node)

Rain outlook

Day	Outlook	Temp.	Humidity	Wind	Decision
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
10	Rain	Mild	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Calculate gini index scores for temperature, humidity and wind features when outlook is rain.

Gini of temprature for rain outlook

Temperature	Yes	No	Number of instances
Cool	1	1	2
Mild	2	1	3

$$\text{Gini(Outlook=Rain and Temp.=Cool)} = 1 - (1/2)^2 - (1/2)^2 = 0.5$$

$$\text{Gini(Outlook=Rain and Temp.=Mild)} = 1 - (2/3)^2 - (1/3)^2 = 0.444$$

$$\text{Gini(Outlook=Rain and Temp.)} = (2/5)\times 0.5 + (3/5)\times 0.444 = 0.466$$

Gini of humidity for rain outlook

Humidity	Yes	No	Number of instances
High	1	1	2
Normal	2	1	3

Gini(Outlook=Rain and Humidity=High) = $1 - (1/2)^2 - (1/2)^2 = 0.5$

Gini(Outlook=Rain and Humidity=Normal) = $1 - (2/3)^2 - (1/3)^2 = 0.444$

Gini(Outlook=Rain and Humidity) = $(2/5)\times0.5 + (3/5)\times0.444 = 0.466$

Gini of wind for rain outlook

Wind	Yes	No	Number of instances
Weak	3	0	3
Strong	0	2	2

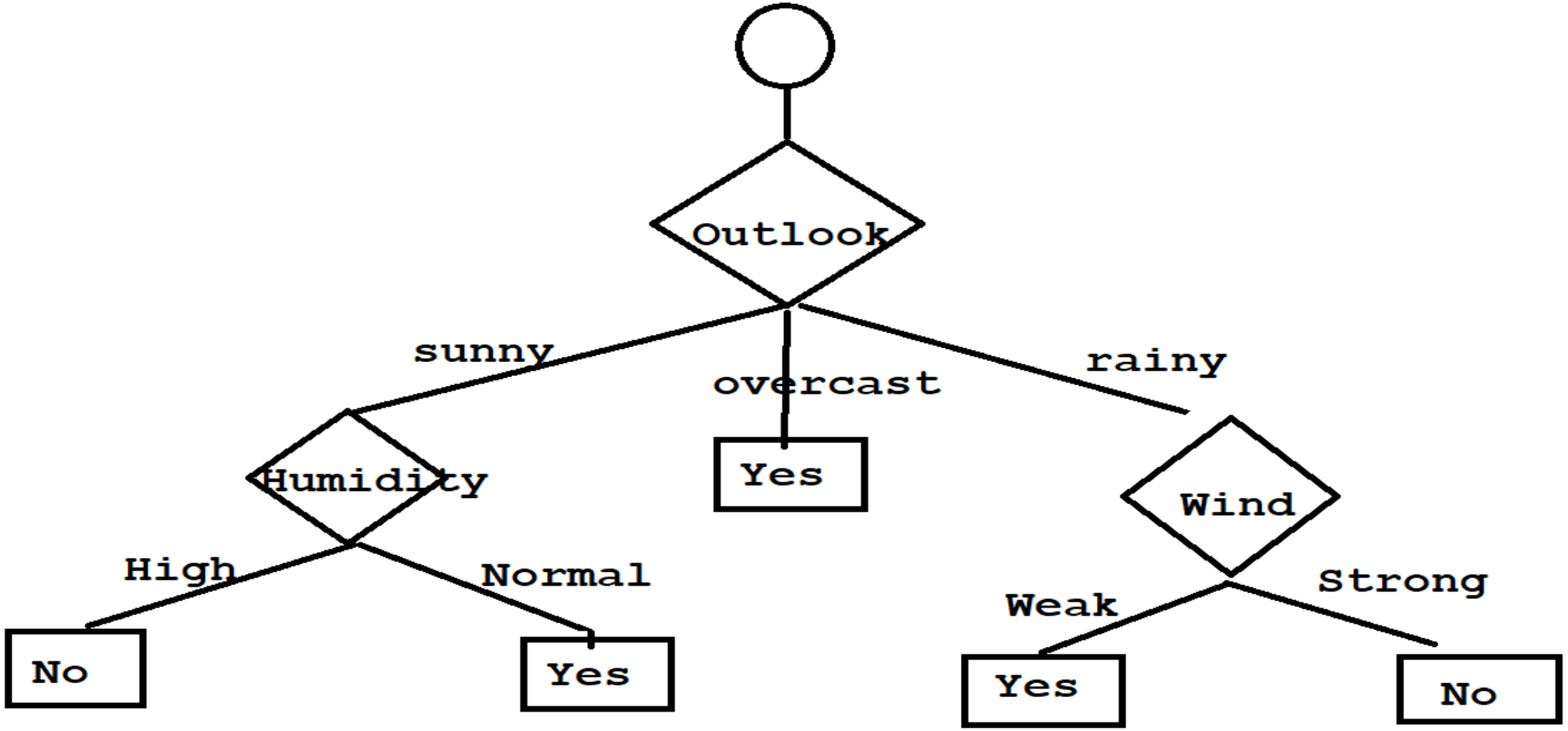
Gini(Outlook=Rain and Wind=Weak) = $1 - (3/3)^2 - (0/3)^2 = 0$

Gini(Outlook=Rain and Wind=Strong) = $1 - (0/2)^2 - (2/2)^2 = 0$

Gini(Outlook=Rain and Wind) = $(3/5)x0 + (2/5)x0 = 0$

The Node under rain will be selected as Wind because among temperature, humidity and wind, wind is having the lowest gini index.

Feature	Gini index
Temperature	0.466
Humidity	0.466
Wind	0



All records with wind Weak, the decision is “Yes”
All records with wind strong, the decision is “No”
So no need to calculate further.

So for a new record

Outlook	Temperature	Humidity	Wind	Play Tennis
Sunny	Mild	High	Weak	?

So according to the decision tree created the result will be

“No”

There will be no need to check for Temperature and Wind at all.